1.7 Equilibria

Question Paper

Course	CIE A Level Chemistry
Section	1. Physical Chemistry
Topic	1.7 Equilibria
Difficulty	Hard

Time allowed: 20

Score: /10

Percentage: /100

Question 1

Use of the periodic table is relevant to this question.

A sample of potassium oxide, K_2O, is dissolved in 500 cm^3 of distilled water. 25.0 cm^3 of this solution is titrated against sulfuric acid of concentration 1.5 mol dm^{-3}. 20.0 cm^3 of this sulfuric acid is needed for complete neutralisation.

What is the mass of potassium oxide was originally dissolved in 500 cm^3 of distilled water?

A. 2.83 g

B. 28.3 g

C. 47.1 g

D. 56.6 g

[1 mark]

Question 2

A dynamic equilibrium was set up between carbon monoxide hydrogen and methanol in a 2 dm^3 sealed vessel at a constant temperature.

$$CO\ (g) + 2H_2\ (g) \rightleftharpoons CH_3OH\ (g)$$

The number of moles of each gas at equilibrium was, carbon monoxide 6.20×10^{-3}, hydrogen 4.80×10^{-3}, methanol 5.20×10^{-5}.

Where has the equilibrium shifted, and what is the value for K_c at this temperature?

A. To the right, $K_c = 364$

B. To the left, $K_c = 364$

C. To the left, $K_c = 1456.1$

D. To the right, $K_c = 1456.1$

[1 mark]

Question 3

Gaseous reactants A and B react to form a gaseous product C in a 1: 3: 2 ratio in a 5 dm^3 container. In an experiment, 1.0 mol of A and 1.0 mol of B are used, and the reaction is heated until equilibrium is reached. At equilibrium, only 10% of A had reacted.

What is the equilibrium concentration of product C at this temperature?

A. 0.18 mol dm^{-3}

B. 0.02 mol dm^{-3}

C. 0.04 mol dm^{-3}

D. 0.2 mol dm^{-3}

[1 mark]

Question 4

Four molecules of oxygen can bind to one molecule of haemoglobin, Hb, according to the following equation.

$$Hb\,(aq) + 4O_2\,(aq) \rightleftharpoons Hb\,(O_2)_4\,(aq)$$

At equilibrium, the concentration of oxygen is 6.7×10^{-6} mol dm^{-3} and the equilibrium concentrations of Hb and $Hb(O_2)_4$ are the same.

What is the value of K_c for this equilibrium?

A. 5×10^{20} mol^{-4} dm^{12}

B. 1.3×10^{5} mol^{-4} dm^{12}

C. 7.6×10^{-6} mol^{-4} dm^{12}

D. 3.3×10^{-21} mol^{-4} dm^{12}

[1 mark]

Question 5

An aqueous solution was prepared containing 1.0 mol of silver nitrate and 1.0 mol of iron sulphate (II) in a 2.00 dm^3 container. When equilibrium was established, there was a 0.44 mol of silver ions in the mixture.

$$Ag^+(aq) + Fe^{2+} \rightleftharpoons Ag(s) + Fe^{3+}(aq)$$

What is the numerical value of K_c?

A. 5.79

B. 5.79 $mol^{-1}\ dm^3$

C. 1.27 $mol^{-1}\ dm^3$

D. 1.27

[1 mark]

Question 6

560 kg of nitrogen and 120 kg of hydrogen are pressurized, heated and passed over an iron catalyst. When the mixture of gases reaches equilibrium, it contains 96 kg of hydrogen.

What mass of ammonia will it contain?

A. 1.36×10^2

B. 2.04×10^5

C. 1.36×10^5

D. 2.72×10^5

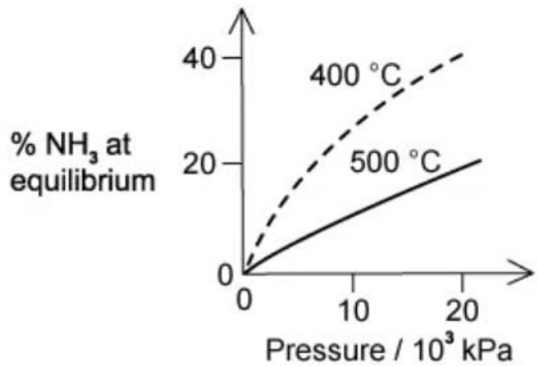
[1 mark]

Question 7

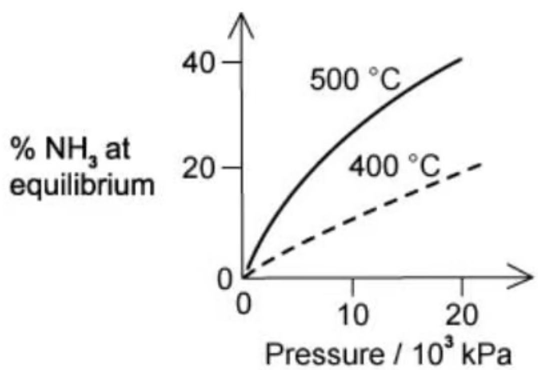
The percentage yield of ammonia produced if the equilibrium was established during the Haber process is plotted against the operating pressure for two temperatures, 400˚C and 500˚C.

Which diagram correctly represents the two graphs?

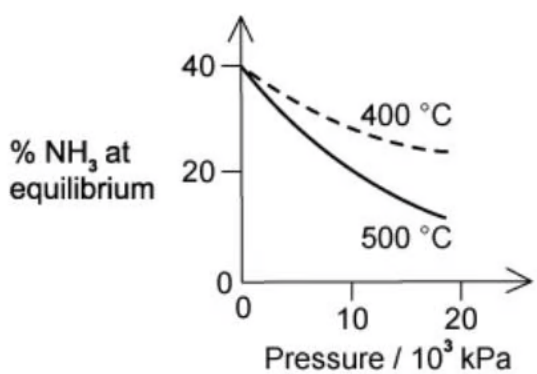
A.



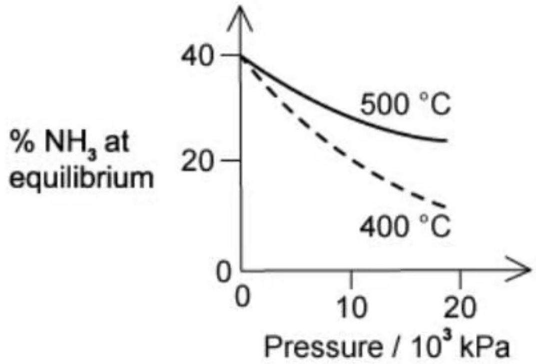
B.



C.



D.



[1 mark]

Question 8

Equimolar quantities of strontium carbonate and magnesium carbonate are heated separately, resulting in complete thermal decomposition. The minimum temperature for the reaction to occur is called T_d.

The cold residues are separately added to equal volumes of water, and the change in pH is measured. The change in pH is called ΔpH.

Which metal has the highest value of T_d, and the greater value of ΔpH?

	T_d	ΔpH
A	Sr	Mg
B	Sr	Sr
C	Mg	Mg
D	Mg	Sr

[1 mark]

Question 9

Shown below are two equilibria.

The numerical value for K_c in reaction I is 2.

Reaction I $2A_2(g) + B_2(g) \rightleftharpoons 2A_2B(g)$

Reaction II $A_2B(g) \rightleftharpoons A_2 + \frac{1}{2}B_2(g)$

Under the same conditions, what is the numerical value of K_c for reaction II?

A. $\frac{1}{2}$

B. $\frac{1}{\sqrt{2}}$

C. 1

D. 4

[1 mark]

Question 10

The following equilibrium is set up in a mixture of sulfuric acid and concentrated nitric acid.

$$H_2SO_4 + HNO_3 \rightleftharpoons HSO_4^- + H_2NO_3^+$$

Which row correctly describes the behaviour of each molecule within the equilibrium?

	H_2SO_4	HNO_3	HSO_4^-	$H_2NO_3^+$
A	Acid	Acid	Base	Base
B	Base	Acid	Acid	Base
C	Acid	Base	Base	Acid
D	Acid	Base	Acid	Base

[1 mark]

Model Answers: Hard

1

The correct answer is **D** because:

- When potassium oxide is dissolved in water, the following reaction takes place:
  - $K_2O + H_2O \rightarrow 2KOH$
  - So one mole of potassium oxide will produce 2 moles of potassium hydroxide.

- The potassium hydroxide is titrated against sulfuric acid in the following reaction:
  - $2KOH + H_2SO_4 \rightarrow K_2SO_4 + 2H_2O$
  - So two moles of potassium hydroxide will react with 1 mole of sulfuric acid.

- **Known quantities:**
  - Concentration sulfuric acid – 1.5 mol dm^{-3}
  - Volume of sulfuric acid – 20 cm^3
  - Moles of sulfuric acid = concentration × volume = $1.5 \times \left(\frac{20}{1000}\right) = 0.03$
  - Volume of potassium hydroxide – 25 cm^3

- One mole of sulfuric acid reacts with 2 moles of potassium hydroxide, so moles of potassium hydroxide = 0.06

- Therefore, in 25 cm^3 there are 0.06 moles of potassium hydroxide, in 500 cm^3 there will be:
  - $0.06 \times \frac{500}{25} = 1.2\ mol$

- The question asks for the mass of the potassium oxide dissolved in 500 cm$^{3.}$

- One mole of potassium oxide makes 2 moles of potassium hydroxide, so the original sample contains 0.6 mol of potassium oxide:
  - mass = moles × M_r = 0.6 × 94 = 56.4 g

- The closest answer is option **D.**

2

The correct answer is **D** because:

- **Divide the equilibrium moles by the volume (2 dm^{-3}) to get the equilibrium concentrations.**
  - [CO] = (6.20×10^{-3}) / 2 = 3.10×10^{-3} mol dm^{-3}
  - [H_2] = (4.80×10^{-3}) / 2 = 2.40×10^{-3} mol dm^{-3}
  - [CH_3OH] = (5.20×10^{-5}) / 2 = 2.60×10^{-5} mol dm^{-3}

- **Write out the K_c equation for this reaction:**
  - $K_c = \frac{[CH_3OH]}{[CO]\ [H_2]^2}$

- **Input the values**
  - $K_c = \frac{(2.6 \times 10^{-5})}{(3.10 \times 10^{-3})\ (2.40 \times 10^{-3})^2} = \mathbf{\mathit{14.56}}$

- As the value for K_c is greater than 1 the equilibrium must be shifted to the right to make the numerator (products) greater than the denominator (reactants).

A is incorrect as whilst the equilibrium has shifted to the right; this incorrect value was obtained by not dividing the equilibrium moles by the volume to get equilibrium concentrations.

B & C is incorrect as for the equilibrium to have shifted to the left, the value of K_c must be less than zero.

3

The correct answer is **C** because:

- Create an initial, change and equilibrium table. Remember to use **the molar ratio of the change.**

- Remember to divide the equilibrium moles by 5 (volume is 5 dm^3) to get the equilibrium concentrations.

- 10 % of A has reacted, 10% of the initial moles of 1.0 is 0.1.

- Therefore 0.1 is the change we use in our table.

	A (g) +	3B (g) ⇌	2C (g)

Initial moles	1.0	1.0	0.0
Change	-0.1	-0.3	+0.2
Equilibrium moles	0.9	0.7	0.2
Equilibrium conc.	0.18	0.14	**0.04**

A is incorrect as this is the equilibrium concentration of A, as seen in the table above.

B is incorrect as this incorrect answer is derived using the incorrect molar ratio of the change. 0.1 is used (this implies a 1 : 1 molar ratio) 0.1 / 5 = 0.02 mol dm^{-3}. The molar ratio of A : B is 1 : 2. Not 1 : 1.

D is incorrect as this is the equilibrium moles, not the concentration. You need to divide this answer by 5 dm^3.

4

The correct answer is **A** because:

- **Write out the expression for K_c**
  - $K_c = \frac{[Hb(O_2)_4]}{[Hb][O_2]^4}$
- **As the concentrations of Hb and $Hb(O_2)_4$ are the same, simplify the equation**
  - $Kc = \frac{1}{[O_2]^4}$
- **Substitute into the equation**
  - $K_c = \frac{1}{(6.7 \times 10^{-6})^4\ mol^4\ dm^{-12}} = 4.96 \times 10^{20}\ mol^{-4}\ dm^{12}$

B is incorrect as the value of $[O_2]$ was not raised to the power of 4. This incorrect answer was derived via $K_c = \frac{1}{[O_2]}$.

C & D are incorrect as these can be ruled out because, with the correct formula for K_c, the value is going to be a very large, greater than 1. The equilibrium has shifted very far to the right. As these are less than 1 they cannot be correct.

5

The correct answer is **B** because:

$[O_2]^4$

- **Substitute into the equation**
  - $K_c = \frac{1}{(6.7 \times 10^{-6})^4 \ mol^4 \ dm^{-12}} = 4.96 \times 10^{20} \ mol^{-4} \ dm^{12}$

B is incorrect as the value of $[O_2]$ was not raised to the power of 4. This incorrect answer was derived via $K_c = \frac{1}{[O_2]}$.

C & D are incorrect as these can be ruled out because, with the correct formula for K_c, the value is going to be a very large, greater than 1. The equilibrium has shifted very far to the right. As these are less than 1 they cannot be correct.

5

The correct answer is **B** because:

- Create an initial, change and equilibrium table. Remember to use **the molar ratio of the change.**
- Remember to divide the equilibrium moles by 2 (volume is 2 dm^3) to get the equilibrium concentrations.
- Solids are not included in *K*c as its concentration is constant.

	Ag+ (aq) +	Fe^{2+} (aq) $\rightleftharpoons$	Fe^{3+} (aq)
Initial moles	1.0	1.0	0.0
Change	-0.56	-0.56	+0.56
Equilibrium moles	0.44	0.44	0.56
Equilibrium conc.	0.22	0.22	0.28

- **Write out the expression for K_c**
  - $K_c = \frac{[Fe^{3+}] \ mol \ dm^{-3}}{[Ag^+] \ mol \ dm^{-3} \ [Fe^{2+}] \ mol \ dm^{-3}}$

- **Substitute into the equation**
  - $K_c = \dfrac{0.28\ mol\ dm^{-3}}{0.22^2\ mol^2\ dm^{-6}} = 5.79\ mol^{-1}\ dm^3$

A & **D** are incorrect as the units are incorrect. Solids are not included in the equation for K_c.

C is incorrect as the value for the Ag^+ (aq) has not been squared. When you have the same molar ratio and starting moles, you will have the same equilibrium moles and thus equilibrium concentrations.

6

The correct answer is **C** because:

- **Convert all masses from kg into g by multiplying by 1000.**
  - N_2 = 560 x 1000 = 560 000 g
  - H_2 = 120 x 1000 = 120 000 g
  - H_2 at equilibrium = 96 x 1000 = 96 000 g

- **Calculate the moles of each using** $\textbf{moles} = \dfrac{\textbf{mass}}{\textbf{molar mass}}$
  - N_2 = 560 000 / 28 = 20 000 mol
  - H_2 = 120 000 / 2 = 60 000 mol
  - H_2 at equilibrium = 48 000 mol

- **Create an initial, change and equilibrium table**
  - Remember to use **the molar ratio of the change.**
  - **Hint:** You only need to fill out the hydrogen and ammonia columns to get the answer.

	N_2 (g) +	$3H_2$ (g) ⇌	$2NH_3$ (g)
Initial moles	20 000	60 000	0
Change	- 4,000	- 12,000	+ 8,000
Equilibrium moles	16,000	48,000	**8,000**

- **Calculate the mass of ammonia**

- mass = moles × molar mass
- Mass of NH_3 at equilibrium = 8 000 mol x 17 g mol^{-1} = **1.36 x 10^5 g**

A is incorrect as the units were not converted from kg into g.

B is incorrect as the incorrect equilibrium moles was used for NH_3 (12 000 moles). This would only be the case if the molar ratio of H_2 : NH_3 was 1 : 1.

D is incorrect as the incorrect molar mass was used (2 x molar mass of NH_3).

7

The correct answer is **A** because:

- The Haber process is:
  - $N_2 (g) + 3H_2 (g) \rightleftharpoons 2NH_3 (g)$
- When the temperature increases the kinetic energy of the particles will increase, increasing the chances of successful collisions in a reaction.
- This reaction is exothermic so when the temperature increases it will favour the backwards endothermic reaction.
- As pressure increases the reaction constant decreases as there are 2 moles of gas produced from 4 moles of reactants.
- The reaction will favour the forward reaction at high pressures, and more ammonia is formed.
- The correct option is, therefore graph **A.**
- This shows a decreased production of ammonia at high temperatures and an increase of ammonia at a higher pressure.

B & **D** are incorrect as these graphs show an increase in production at higher temperatures.

C is incorrect as this shows a reduction in the production of ammonia at a higher pressure.

8

The correct answer is **B** because:

- All the group 2 metals undergo thermal decomposition to produce the metal oxide

- H_2 = 120 000 / 2 = 60 000 mol
- H_2 at equilibrium = 48 000 mol

- **Create an initial, change and equilibrium table**
  - Remember to use **the molar ratio of the change.**
  - **Hint:** You only need to fill out the hydrogen and ammonia columns to get the answer.

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  - mass = moles × molar mass
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B & **D** are incorrect as these graphs show an increase in production at higher temperatures.

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8

The correct answer is **B** because:

- All the group 2 metals undergo thermal decomposition to produce the metal oxide and carbon dioxide gas.
  - XCO_3 (s) → XO (s) + CO_2 (g)
  - Where X represents the group 2 metal.
- As you go down the group, more energy is needed to decompose the compound thermally.
  - **They are more stable as you go down the group.**
- Strontium is below magnesium and will have a higher T_d than magnesium.
- The reaction of the metal oxide with water will produce a metal hydroxide.
  - XO (s) + H_2O (l) → $X(OH)_2$ (aq)
- Strontium hydroxide is more soluble than magnesium hydroxide.
- Therefore it will have a higher ΔpH as the more soluble the hydroxide, the more alkaline.
  - **If a substance is more soluble there will be more OH^- ions available per mole of metal hydroxide dissolved.**
  - **This means that the solution will be more alkaline and therefore have a higher**

pH.

9

The correct answer is **B** because:

- **Reaction I expression for Kc**
  - $K_{cI} = \dfrac{[A_2B]^2}{[A_2]^2[B_2]}$
- **Reaction II expression for *Kc***
  - $K_{cII} = \dfrac{[B_2]^{\frac{1}{2}}[A_2]}{[A_2B]}$
- **Reaction I** & **II** run in the opposite directions. If the K_c of **reaction I** is 2 then in **reaction II** the reciprocal of K_c is taken to $K_c = ½$.
- In **reaction I** there are 2 moles of reactants and products so would be squared. In **reaction II** it is just one mole, we would therefore need to take the square root of each side hence $\dfrac{1}{\sqrt{2}}$ (option **B).**

10

The correct answer is **C** because:

- Neutralisation reactions occur when H^+ (aq) and OH^- (aq) form H_2O (l) and salts.
- The Brønsted-Lowry definition of acids and bases as proton donors and acceptors.
  - A Brønsted-Lowry **acid is an H^+ donor**
  - A Brønsted-Lowry **base is an H^+ acceptor**
- **Forward reaction**
  - Sulfuric acid is acting as an acid and donating an H^+ ion
  - Nitric acid is acting as a base as it is accepting an H^+ ion
- **Backwards reaction**

- HSO_4^- is acting as a base, accepting an H^+ ion
- $H_2NO_3^+$ is acting as an acid, donating an H^+ ion